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Optimizing Horizontal Open Hole Completion: Harnessing Technical Limit to Significantly Reduce Total Rig Time of Limited Entry Liner Installation

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Abstract

This paper presents a methodology for employing the technical limit approach to minimize invisible lost time and optimize lower completion installation. The focus is on enhancing the completion fluid recipe and eliminating unnecessary operational steps, leading to significant reduction in rig time.

Running a lower completion string with up to 7 swellable packers of 5 3/4" OD through a 6 1/8" horizontal hole section of ±600 m length was challenging. Several engineering and operations modifications were applied to overcome the challenges and reduce the total rig time. The lower completion string was run in the wellbore with a viscous, lubricated brine, other than drilling fluids, which allowed for the discard of the inner string in the limited entry liner (LEL). A pre-rack pack of drill pipes helps with running in hole completion by stands, other than pickup and RIH, using single joints.

The engineering and operational optimization proposed to optimize the deployment of the LEL assembly were successfully implemented in Well B, with fewer risks and a shorter time frame compared to Well A. In Well A, the LEL ran while drilling mud was in the hole, and the inner string was utilized to allow for circulating and completion fluid displacement. The primary disadvantage of using the inner string is that it causes the LEL assembly to operate with higher drag forces. To eliminate the inner string, the liner hanger setting configuration was modified by adding a Rotational Ball Seat Assembly (RBSA) and a ball catcher combined with hanger running tool. Since circulating is not an option without an inner string, a brine fluid was used instead of drilling mud to run the lower completion. In the lab, various tests were conducted to achieve the desired brine rheology and lubricity properties. Eventually, a viscous, lubricated brine was introduced. A drill pipe's pre-rack stands were installed to run the LEL to the target depth, minimizing stationary time across a depleted formation. The total rig time for well A to run and install the LEL was 15 days, compared with 9 days in Well B, resulting in a 42% reduction in total rig time.

This work aims to share a successful optimization process for installation the limited- entry liner in two horizontal wells. It provides an overview of employing the technical limit approach in engineering and operations to reduce total rig time by 42%. Operation optimization and fluid recipe modification are now being adopted for current and future lower completion installations and are expected to have a positive impact on similar projects.

Introduction

LEL concept is not something new in the industry. The concept was proposed in the early 1960's by Shell for the hydraulic fracturing application [Talib N. 2021](#). However, the installation of an open-hole lower completion in a horizontal well is a relatively new scope in the Iraq oil fields. In a giant oil field, where many horizontal wells have been drilled successfully, a new challenge was encountered which is installing the open-hole lower completion of LEL.

Mitigating operational risks and improving preperformance is the ultimate goal of all the oil and gas industry entities. Technical limit is a term used to describe a level of performance defined as the best possible for a given set of design parameters [Bond, 1996](#). Among several lower completion installations, a specific well was chosen to demonstrate how the technical limit application can significantly improve performance and reduce rig operations time. The lower completion was planned to run into a horizontal section of ~600 m in two wells, A and B, with a slotted liner string and swellable packers in between. Several initiatives were undertaken to enhance the engineering design and operational procedures, thereby mitigating risks and improving performance.

To achieve the technical limit objective, the offset wells operation steps has been reviewed and the unnecessary steps were removed. The run-in hole of the liner string was utilized by pre-racked pack drill pipes, and the polished mill assembly to dress the Tie-back Receptacle (TBR) was removed. Both initiatives saved many working hours without impacting the quality of the completion process.

From the engineering side, the liner hanger configurations have been modified, where the inner string was replaced with a different tool, combined with a running tool as an alternative. As a result of removing the inner string, the advantage of having circulation through the string is lost. Eventually, the LEL should be run through a completion fluid, but this fluid should possess the required rheological properties to enable the LEL to reach its target properly. A brine of 1.08 SG CaCl₂ was used with additives for viscosity and rheology enhancements.

Limited Entry Liner Objective

LEL implementation strategy is one of the key solutions to improve the well productivity by maximizing the reservoir contact and matrix stimulation in carbonate reservoirs [Talib N. 2021](#). The limited entry liner is a non-cemented liner hanger where normally deployed in a horizontal well section connected with slotted (or pre-perforated) liner joints and swellable packers in between ([Figure 1](#)— limited entry liner schematic). The main objective is to have a mechanical acid diverter where the swellable packer compartments are designed based on the reservoir rocks quality and the vertical permeability distribution across the horizontal section. The more advanced design can be utilized with a smart completion with Inflow Control Valves (ICV) where the flow can be controlled from the surface from the field operation office. It was verified that the LEL deployment made a good improvement on production after performing acid stimulation by a coiled tubing where the LEL functioned as a mechanical acid diverter. The LEL with swellable packers in place ensures that the acid distributed more evenly among a different formation rocks quality.

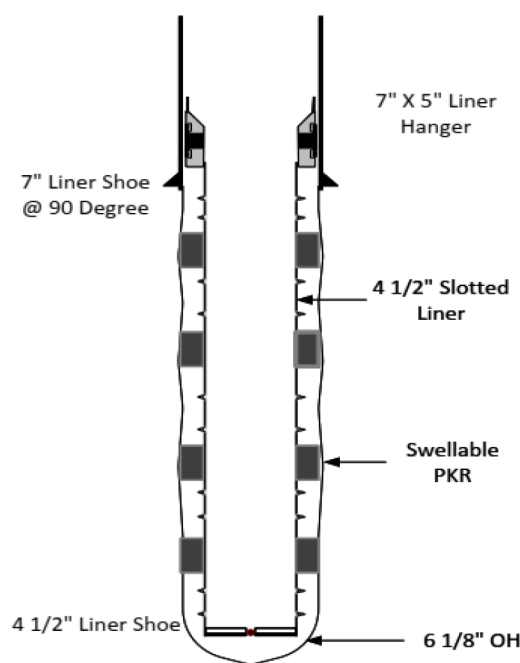


Figure 1—Limited entry liner schematic

Offset Well A (Base Case)

The base case (well A) took about 15 days to retrieve the old upper completion and install the lower completion (LEL) with a Tie-back Seal Assembly (TBSA) as an upper completion. The LEL consists of a 7" × 5" liner hanger and a 4 1/2" slotted liner with seven 5 3/4" swellable packers in between. The total LEL length was approximately 900 m, and it was run to a depth of 3,370 m, where the lateral section lengths were approximately 600 m. During the run-in hole and installation of the LEL completion, several challenges were faced in this well. Starting from severe losses while in the reservoir horizontal section, maintaining the completion assembly was essential to keep the well fluid barrier intact. The drag forces while RIH were high; several times, the string hung up due to tight spots and high friction and drag.

LEL design and Liner Hanger Configuration

The lower completion string consists of 7 swellable packers (5 3/4" OD) to build 6 compartments, where a slotted liner string of 4 1/2" OD casing joints is placed between every two packers. The number and size of slots (perfs) are designed based on the permeability distribution across the formation, in addition to other rock properties. Ultimately, it depends on the quality and heterogeneity of the reservoir rocks. The liner hanger is a conventional design featuring hanger slips that are set hydraulically and the liner top packer that is set mechanically by weight. The inner string connects to the end of the hanger running tool exactly connected to the swivel item (where if required to rotate the hanger, the inner string will not be rotated). Refer to the diagram below, which shows the comment on the lower completion string along with the liner hanger running tool. (The red shaded area shows the pressure)

Inner string and liner setting

Having the circulating option while running the slotted liner is essential for overcoming any tight spots and ledges, especially when navigating through a deviated and horizontal section. At this stage, the inner string function is the only tool available in the project that capable of achieving this purpose, where the inner string will bypass the slotted liner and create a seal once it is inserted into the O-ring seals in the main body of the liner string. Furthermore, the inner string will create a closed system, allowing pressure increase to set the liner hanger hydraulically. Once the liner setting ball dropped through the drill pipe, passing the hanger

and inner string until it landed in the flow isolation valve (FIV), as shown in Fig. 2, the schematic of the closed system during liner hanger setting.

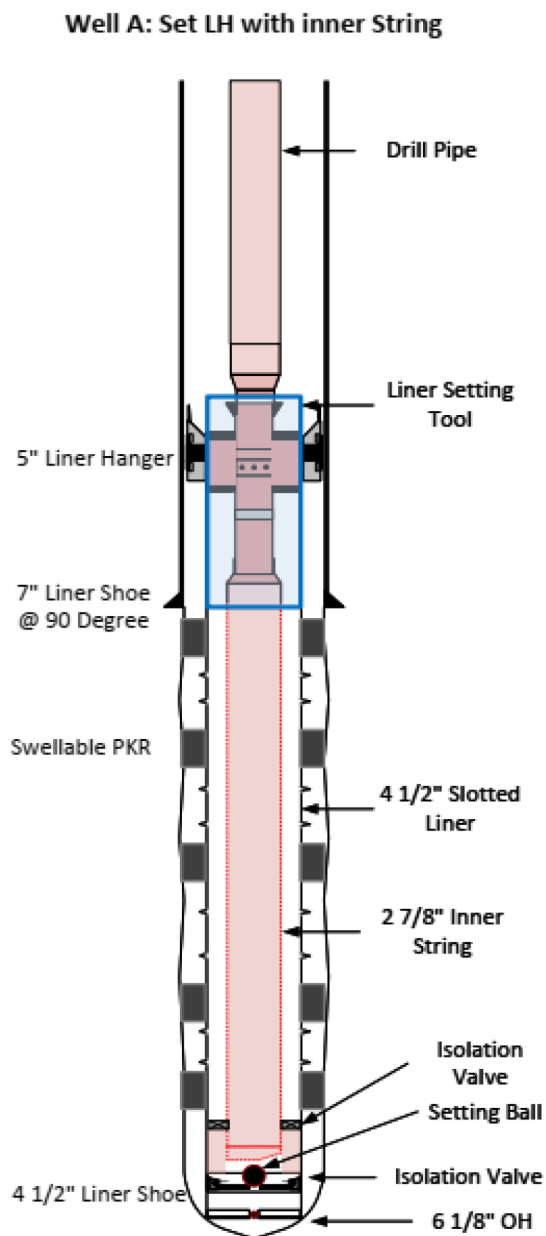


Figure 2—Limited entry liner with inner string (While setting the LH)

The disadvantage of such a setting mechanism is that the ball will travel a long distance from the surface to the shoe track of the liner string passing through different sizes of tubulars (Drill pipe, crossovers, Hanger, and inner string). There is an extra risk of not having the ball reach its destination, which is the high inclination of the wellbore where the landing position is at 90 degrees. In addition, the long distance the ball travels through different tubular IDs will increase the risk of the ball wearing, which will consequently lead to not having built-up pressure while setting the Hanger due to incomplete seals by the ball.

Completion Fluids

There are two main fluids used in this case: the 1.05 SG reservoir-friendly drilling mud and the NaCl brine. The drilling mud objective here was not to drill where the well had already been drilled. The wellbore was

displaced with drilling mud by a dedicated clean-out BHA. The mud functions help create a filter cake that minimizes losses and the risk of differential sticking. The LEL lower completion run was performed through the mud until it reached the liner hanger setting depth. After setting the liner hanger and shearing the ball, the mud was displaced by an inhabited completion brine.

Mud was displaced once the lower completion string reached its designed target. The inner string plays a vital role in allowing circulation through the slotted liner string and displaces the mud with 1.05 SG CaCl₂ brine.

LEL Drag Simulation with Inner String

The combined strings of 4 ½" casing joints with the inner string of 2 7/8" tubing resulted in higher drag while tripping. The simulation of hook loads for RIH & POOH, as shown in the Figure 3—, reveals a considerably higher load of up to 110 tons compared to the optimized case (Well B).

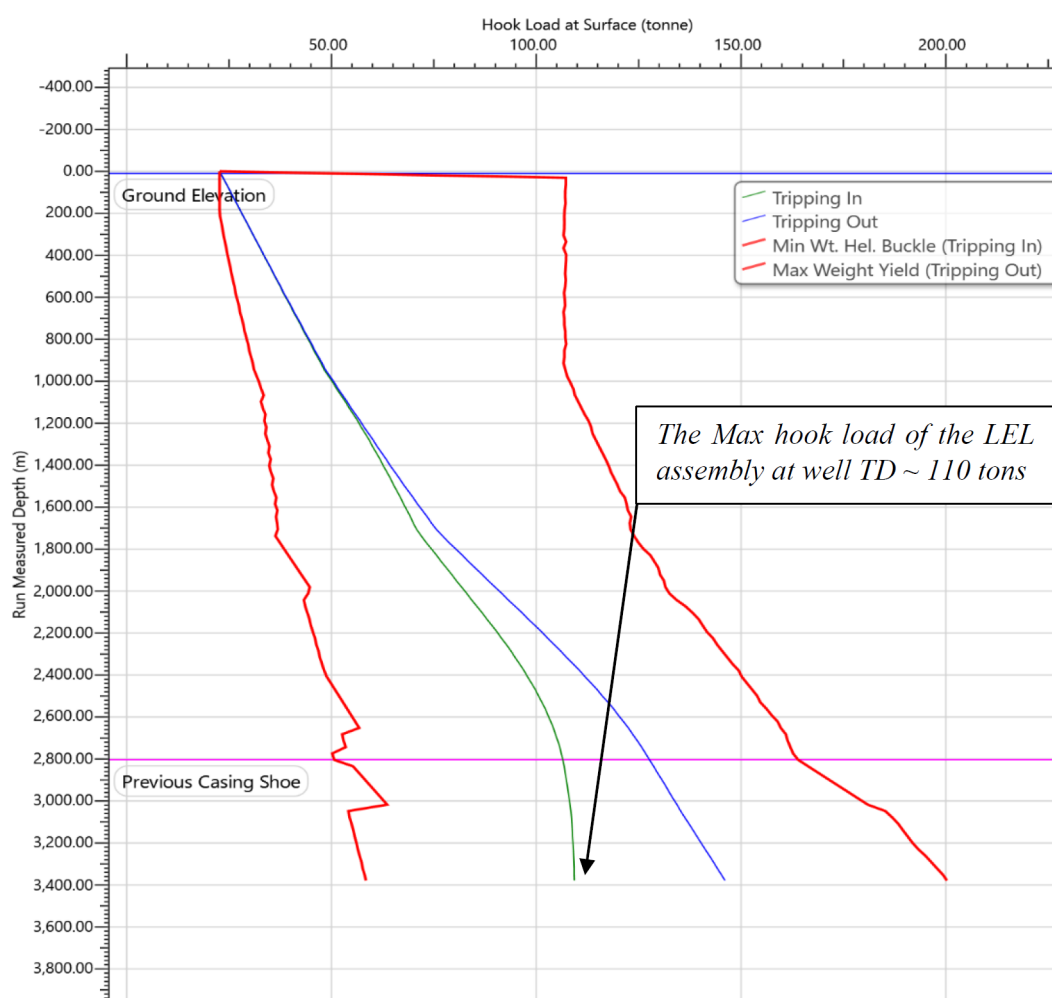


Figure 3—Base case (Well A) LEL tripping drag simulation

Operations Time Breakdown

It is essential for companies not to target performance for the first few wells or at the beginning of an oil and gas project. Instead, they are recommended to allow the crew to build their learning curve, ideally over at least three months or two drilled wells. When examining the time consumed in detail for any scope of work, those with decent previous experience can identify the Invisible Lost Time (ILT). Companies put an

enormous effort into removing such lost time; consequently, in this case, the efforts were focused on the possible engineering modifications and the necessity of some operational steps.

Several challenges were encountered during the LEL installations. Starting from the well, the kill was not properly executed due to a low flow rate, which led to an extended flow check time. Breaking the containment did not go as planned due to an extended flow check time, resulting in a 19-hour completion of this process. POOH and RIH, the tubing did not meet the KPIs of the drilling rig, as the crew is still at the beginning of building the learning curve.

Tripping Time and Challenges

Running the clean-out BHA took considerable time, as the process involved picking up the single drill pipe joints. Having spent more than 5 days on the clean-out BHA, it is something that needs to be optimized. The LEL completion took considerable time as well, during which extra time was spent on RIH and POOH of the inner string. This combination made the string heavier, resulting in a more cautious and slower RIH speed to avoid surge pressure and the risk of sticking.

Table 1—Base Case (Well A) Operations Time Breakdown

Operation Phase	Well A: Time (hrs.)
kill the well	15
N/D X-Tree and N/U BOP	19
Retrieve Upper completion string	46
Cleanout Run	134
Run Lower Completion to TD	94
Run Upper Completion String	44
N/D BOP and N/U X-Tree	10
Total Time (hrs.)	362
Total Time (Days)	15.0

Based on the analysis of operational sequences, tools, and the engineering plan, there is room for improvement. A proposed optimization and improvement to reduce the risks and improve the performance of deployment to save time. The following suggested changes should be implemented in the upcoming wells.

Proposed Optimizations to Reduce the risk and time

1. Rig up the well kill surface lines and pressure test before spud the well.
2. Rack back the required drill pipes on the rig floor before spudding the well.
3. Optimize the well kill (Bullhead) flow rate to exceed the gas migration velocity.
4. Modify the liner hanger configuration to discard the inner string.
5. Optimize the well completion fluid with enough swelling time
6. Discard the slick line plug installation and pressure test the tubing from the annulus side.

Well B (Optimized Case)

The approach of technical limits has been in the oil and gas industry for decades. It seeks alternative methodologies, tools, and techniques to achieve the best possible performance for a specific operation, taking into account the available technology and personnel. In the optimized case, we were focused on three possible changes to deliver better LEL installations.

1. Discard unnecessary operation steps and improve operational performance.
 - o Rack back the required drill pipes before spudding the well to run the LEL in a stand.
 - o Discard the TOL dress mill BHA before running the TBSA, as there are no drilling and cementing operations, so the chance of having debris and cuttings at the TOL is very low.
 - o Keep the slick line plug as a contingency when the pressure test of the tubing from the annulus fails.
2. Modify the liner hanger running tool assembly to remove the inner string.
3. Adopt an alternative fluid to run the LEL through a completion fluid instead of using the drilling mud.

After applying the above main changes, Well B took about 9 days to retrieve the old upper completion and install the LEL with a TBSA as an upper completion. The lower completion assembly installed in this case is nearly identical to that used in Well A, except that it does not include an inner string.

Optimize Liner Hanger Configuration and Design

Running the inner string will bring a risk of installing a false rotary table above the liner hanger's last joints will make the rig floor crew job difficult where the elevation now is higher. Passing the inner string through these joints carries a risk of contact with the box, which may damage the thread.

Due to all the above, removing the inner string will prevent many operational risks and will save time and cost.

The alternative liner hanger design will feature a different tool used to land the ball and create the required pressure build for the hanger setting hydraulically. The Rotation Ball Seat Assembly will be connected at the lowest part of the hanger running tool. This arrangement will eliminate the need for an inner string to set the hanger. However, therefore, the system will lose the option of circulating while RIH or at TD due to the slotted liner joints.

In such a case, running the LEL through completion fluid is the only option available. To do so, several lab tests are conducted to prepare the appropriate completion fluid with the desired rheological properties and lubrication.

This configuration, in addition to the above, offers an advantage that the liner string will not be pressured by the setting the liner hanger as happened in the base case design. The closed system will be between the surface and the liner running tool, where the ball will land.

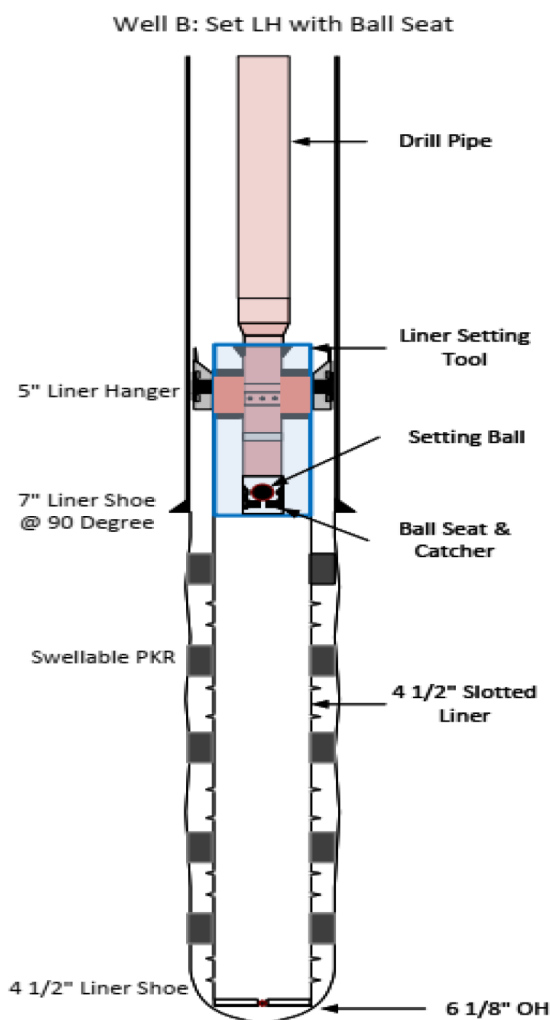


Figure 4—limited entry liner with Ball Seat (While setting the LH)

Optimize Completion fluid recipe

The decision to incorporate a brine dispersible and compatible lubricant into the formulation proved beneficial in facilitating the efficient deployment of the lower completion into the horizontal section [AL Hosni F. 2024](#). There were two types of fluids used in the base case Well A: the drilling mud and the completion brine. In the optimized case, we do not use the drilling mud. That will save costs and minimize the environmental impact in the field area. After numerous trials in the mud lab to determine the optimal brine additive recipes with varying concentrations of lubricant and viscofier additives, the percentage of 1% lubricant and 1-2 ppb viscofier (Polymer) yielded the appropriate rheological properties. Furthermore, the samples were heated to 49 °C to see the effect of the well formation temperature, in addition to leaving the sample for 4 days. The phenomenon of greasing and cheesing is well-known to occur when adding a lubricant; however, there were no signs of a freezing effect or traces of cheesing in the samples, as presented in [Fig. 5](#).

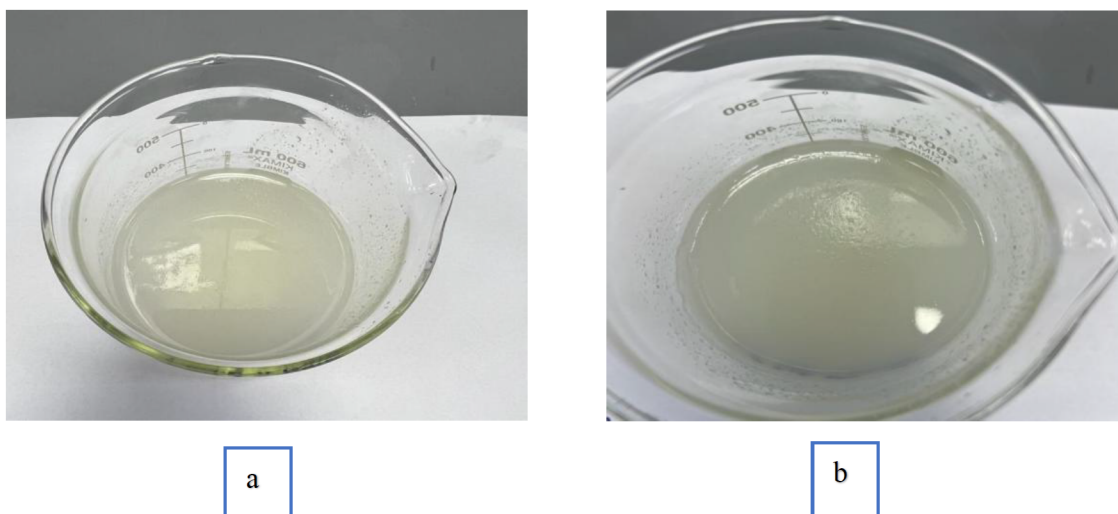


Figure 5—a. Sample Appearance at the time of mixing the additives. b. After 96 Hours: No greasing or cheesing observed with viscosified brine.

Table 2—Rheological readings of brine with additives at Room Temperature, 49 C and after 96 Hours.

	A	B	C
RPM	RT with 1% lube	49 C with 1% lube	RT After 96 Hours
3	6	5	6
6	7	7	8
100	9	14	17
200	12	17	23
300	20	19	27
600	21	20	36

After waiting for four days, the rheological properties of the brine increased when comparing column, A with column C. The most important thing is that there are no signs of greasing and cheesing affecting, meaning the fluid recipe is appropriate and compatible with the designed job.

Lower Completion Tripping Drag

After waiting for four days, the rheological properties of the brine increased when comparing column, A with column C. The most important thing is that there are no signs of greasing and cheesing affecting, meaning the fluid recipe is appropriate and compatible with the designed job. Many lubrication products and hardware tools in the oil and gas industry can be used to minimize the drag on the RIH strings in deviated and horizontal wells. However, optimization does not always mean using more costly and fancy tools to achieve the project goals faster. The collaborations between all relevant personnel successfully led to lower completion rates reaching the targets with fewer complications and no advanced technology or complicated tools. Several factors impact whether the LEL reaches the target.

- Clean out the wellbore, Run, and circulate well to ensure no debris
- RIH stands to minimize the stationary time across a risky formation to avoid differential sticking.
- Removing the inner string reduces the drag forces on the liner string.
- The completion brine plays a vital role in reducing the tripping friction factor.

LEL Drag Simulation without Inner String

The tripping hook load reduced by 22 tons compared with the base case well, as shown in Fig. 6—LEL tripping drag simulation, presented the loads cab be seen at surface while RIH to the section TD. The max load of the assembly for RIH was about 88 tons.

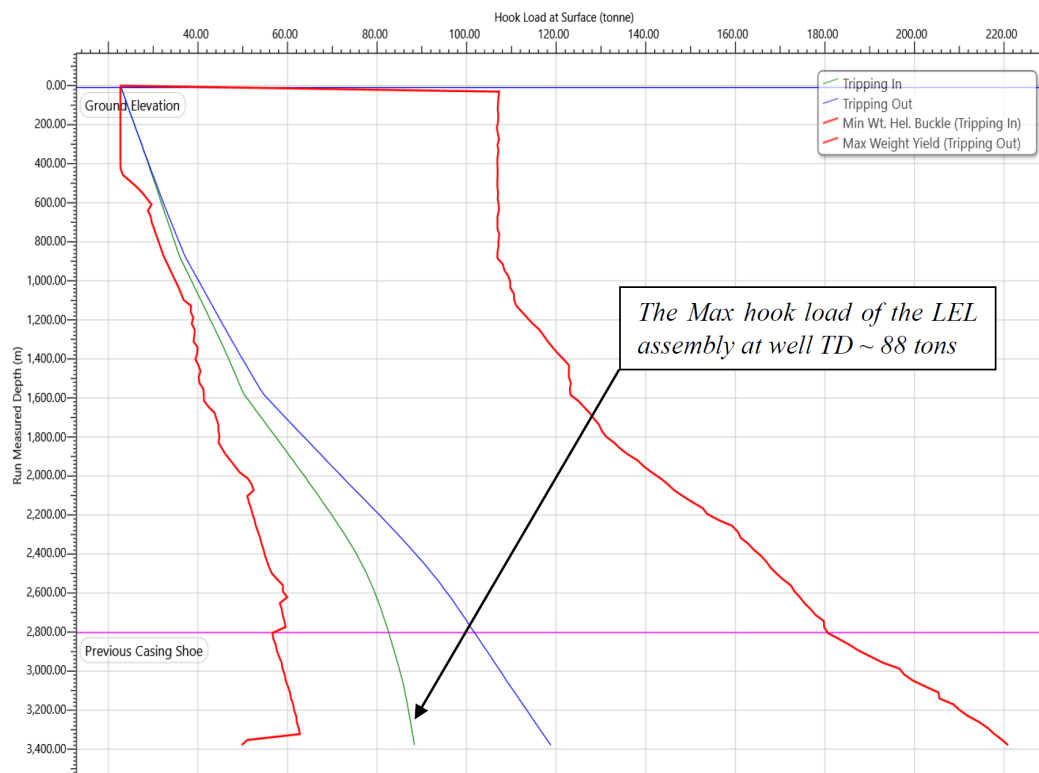


Figure 6—Optimize Case (Well B) LEL tripping drag simulation

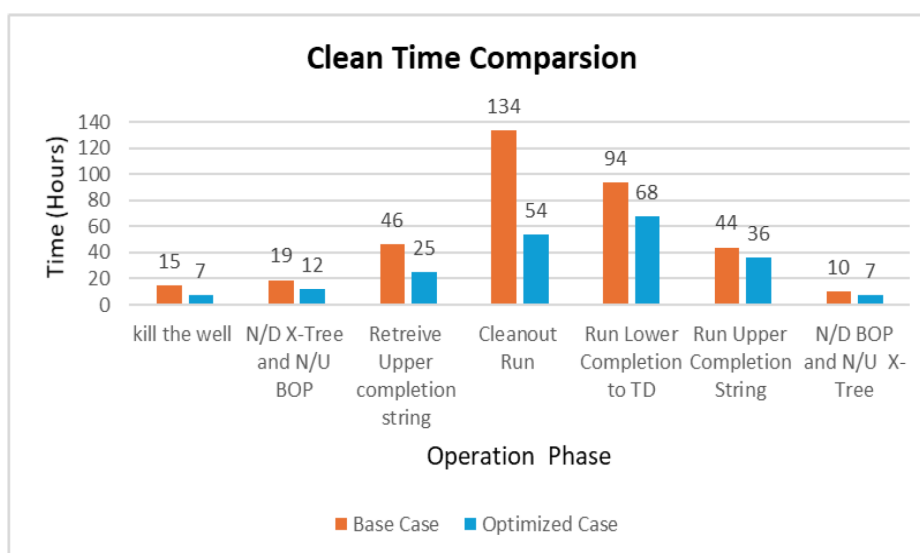


Figure 7—Bar chart clean time comparison

Table 3—Clean time comparisons and improvement percentage

Operation Phase	Well A: Time (hrs.)	Well B: Time (hrs.)	Improvement%	Justification
kill the well	15	7	53%	Being prepared ahead of time. Bullhead process improvement
N/D X-Tree and N/U BOP	19	12	37%	Being prepared ahead of time and Crew performance improvement.
Retrieve Upper completion string	46	25	46%	Crew performance improvement
Cleanout Run	134	54	60%	Run DPs in stands already racked packed before spud
Run Lower Completion to TD	94	68	28%	Remove the inner string and Run DPs in stands. No displacement.
Run Upper Completion String	44	36	18%	Remove the SL run, Crew performance improvement
N/D BOP and N/U X-Tree	10	7	30%	Crew performance improvement
Total Time (Hrs.)	362	209	42%	
Total Time (Days)	15.0	8.7		

Conclusion

Several engineering and operational steps were optimized to achieve a safe and higher-performance LEL and upper completion installation, resulting in approximately a 42% reduction in rig time, as shown in [fig. 7](#)—bar chart and [Table 3](#)—clean time comparisons and improvement percentage. The ability to recognize Invisible Lost Time and implement the appropriate and safe plans to remove it is a significant sign of applying the technical limit approach correctly.

- In Well B, the surface lines were lined up and pressure tested before spudding the well. In addition, the well kill flow rate increased to exceed the gas migration rate, eliminating the need for an extended flow check, as was the case in Well A.
- The breaking of containment for well B was done smoothly, as the well was killed properly, and preparations made ahead of time saved time.
- In well B, the decompletion time improved where the POOH speed optimized and the crew performance affected the operation time as well.
- The DPs were racked back before the well spud, where in Well B, the clean-out BHA ran with a stand.
- The discarded inner string ran the liner with stands, and no displacement for the completion fluids in well B improved the operations time by 28%.
- The discard of the slick line run, and the crew performance improvements reduce the upper completion installation by 8 hours in well B. The slick line plug installation to test the tubing string is replaced by testing the annulus against the liner top packer, and having the SL plug as a contingency option.
- The crew saved about 11 hrs. for the breaking containment process in well B.
- The discard of the drilling mud from the process, save the costs and minimize the environmental impact on the field area.

- The drag force in the Well B was reduced by 20% where the hook load reduced to about 88 tons compare with 110 tons in well A due to removal of the inner string and using the lubricated brine as a completion fluid instead of the drilling mud.

Nomenclature

LEL	Limited Entry Liner
BHA	Bottom Hole Assembly
LH	Liner Hanger
LTP	Liner Top Packer
RIH	Run In Hole
MD	Measure Depth
RBSA	Rotational Ball Seat Assembly
T&D	Torque and Drag
SL	Slick Line
L/D	Lay Down
N/D	Nipple Down
N/U	Nipple Up
BOP	Blow Out Preventer
PSI	Pound Per Squar Inche
PPB	Pound Per Barrel
TBR	Tie-back Receptacle
SG	Specific Gravity
OH	Open Hole
CH	Cased Hole
KPI	Key Performance Indicator
TD	Total Depth
OD	Outside Diameter
ID	Inside Diameter
FF	Friction Factor
TBSA	Tie-Back Seal Assembly

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